

Lecture#10

Binomial Distribution

When an experiment has two possible outcomes, success and failure and the experiment is repeated n times independently and the probability P of success of any given trial remains constant from trial to trial, the experiment is known as binomial experiment.

If X is a binomial random variable, then

$$X \sim \text{Binomial}(n, p)$$

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}, \quad x = 0, 1, 2, \dots, n;$$

where n is the number of trials and p is the probability of success.

[$f(x) = P(X=x)$].

Note. $n=1$, i.e., the number of trials is one, the binomial variable (X) follows Bernoulli distribution. For the Bernoulli distribution, if X is a Bernoulli random variable, then

$$X \sim \text{Bernoulli}(p) \text{ or } X \sim \text{Binomial}(1, p)$$

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x} = p^x (1 - p)^{n-x}, \quad x = 0, 1;$$

Prove that,

- (i) Total probability of Bernoulli random variable is 1.
- (ii) Mean of Bernoulli random variable is $E(X) = p$.
- (iii) Variance of Bernoulli random variable is $V(X) = p(1 - p)$.
- (iv) Total probability of binomial random variable is 1.
- (v) Mean of binomial random variable is $E(X) = np$.
- (vi) Variance of binomial random variable is $V(X) = np(1 - p)$.

(Note. $(x + y)^n = \sum_{k=0}^n n_{C_k} x^k y^{n-k}$)

Example: The probability that a patient recovers from a disease is 0.9. What is the probability that an exactly 5 out of the next 7 patients will survive?

Solution:

We assure that the operations are made independently and $p=0.9$ for each of the seven patients.

n=7, p=0.9 1: success (recover), 0: failure (not recover)							
patient1	p2	p3	p4	p5	p6	p7	Total
1	1	0	0	0	1	0	3
0	0	0	0	1	0	0	1
⏟							⋮
Bernoulli							Binomial

So, $X \sim \text{Binomial}(n=7, p=0.9)$

$$\begin{aligned}
 P(X = 5) &= \binom{n}{x} p^x (1 - p)^{n-x} \\
 &= \binom{7}{5} 0.9^5 (1 - 0.9)^{7-5} = 0.1240
 \end{aligned}$$

_How many patients will survive on an average?

$$E(X) = n * p = 7 * 0.9 = 6.3$$

So, 6.3 patients out of 7 patients will survive on an average. That is, 63 patients out of 70 patients will survive on an average.

Example: A traffic control officer reports that 75% of the trucks passing through a check post are from within Dhaka city. What is the probability that at least 3 of the next 5 trucks are from out of the city?

Solution: Let X be the number of trucks that pass through are from out of Dhaka city. The probability of such an event is: $p=1-0.75=0.25$

$$\begin{aligned}
 X &\sim \text{Binom}(n = 5, p = 0.25) \\
 P(X \geq 3) &= P(X = 3) + P(X = 4) + P(X = 5) \\
 &= \sum_{x=3}^5 \binom{n}{x} p^x (1 - p)^{(n-x)}
 \end{aligned}$$

$$= \binom{5}{3} (0.25^3)(0.75^{5-3}) + \binom{5}{4} 0.25^4 0.75^{5-4} + \binom{5}{5} 0.25^5 0.75^{5-5}$$

$$= 0.1035$$

$$\text{Or, } P(X \geq 3) = 1 - P(X = 0) - P(X = 1) - P(X = 2)$$

$$= 1 - \binom{5}{0} \cdot 25^0 \cdot 75^{5-0} - \binom{5}{1} 0.25^1 0.75^{5-1} - \binom{5}{2} \cdot 25^2 \cdot 75^{5-2}$$

*Probability of more than 3 trucks are out of the city:

$$P(X > 3) = P(X = 4) + P(X = 5)$$

$$= \binom{5}{4} (.25^4)(.75^{5-4}) + \binom{5}{5} (.25^5)(.75^{5-5})$$

$$= 0.0156$$

*Probability that at most 2 trucks are out of the city:

$$P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2) = \binom{5}{0} (0.25)^0 (0.75)^{5-0} + \dots$$

*Probability that less than 2 trucks are out of the city:

$$P(X < 2) = P(X = 0) + P(X = 1)$$

$$*P(1 \leq X < 3) = P(X = 1) + P(X = 2)$$

$$*P(1 < X < 3) = P(X = 2)$$

What is the mean number of trucks out of the Dhaka city passing through the check post?

$$E(X) = n * p = 5 * 0.25 = 1.25$$

So, 125 trucks out of 500 trucks will pass from out of the Dhaka city.

Problem. If $E(X) = 5$, $V(X) = 2$, Then, find P and n .

Given, $E(X) = nP = 5$, and, $V(X) = nP(1 - P) = 2$

$$\frac{nP(1-P)}{nP} = \frac{2}{5}$$

$$\text{or, } 1-P = \frac{2}{5}$$

$$\text{or, } P = 1 - \frac{2}{5} = \frac{3}{5}$$

Now, $nP = 5$

$$\text{or, } n = \frac{5}{P} = \frac{5 \times 5}{3} = \frac{25}{3}$$

$$X \sim \text{Bin}(n = \frac{25}{3}, P = \frac{3}{5})$$

Exercise.

1. You toss a fair coin 10 times. What is the probability of exactly 7 heads?
2. A batch contains many items; each item independently is defective with probability 0.05. You inspect 20 items. Compute the probability that (i) at most one item is defective, (ii) at least three items are defective, and defective items lie between 2 and 5.
3. A clinical trial gives each patient an independent probability 0.60 of responding positively. You treat 15 patients. (a) Find the probability that at least 10 patients respond, (b) Find the mean and variance of responding.